SEE4994
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SYNOPTIC DRIVERS OF HIGH-POLLUTION DAYS IN MEXICO CITY (2012–2024)

Week 6 – 500 hPa ridge anomalies, atmospheric stability, and comparative patterns (CDMX ↔ HK)

- Week 6 builds on the Week 5 composites by quantifying the ridge-stagnation pattern and benchmarking it against published regimes in Mexico and Hong Kong.
- A mid-tropospheric ridge with weak ENE flow is the signature of CDMX peak-pollution days.

What large-scale circulation pattern governs extreme air-pollution days in Mexico City, and how does it compare with Hong Kong's regional meteorology?

WHAT CHANGED IN WEEK 6 (VS. WEEK 5) ?

- Rebuilt composites as anomalies H′ using a 31-day smoothed day-of-year climatology.
- Focused on 500 hPa (closest to MCMA* topography signal; fastest to run robustly).
- Adopted “PhD-style” plot recipe: H′ shading + H′ contours, vectors, significance stippling, MCMA box, lat/lon grid, consistent colormap.
- Event definition kept as monthly p90 (per pollutant, 2012–2024).
- Added diagnostics: mean H′ and 500-hPa winds over MCMA; monthly event counts.
- Key takeaway: Results are now climatology-referenced and directly interpretable synoptically.

*MCMA: Mexico City Metropolitan Area

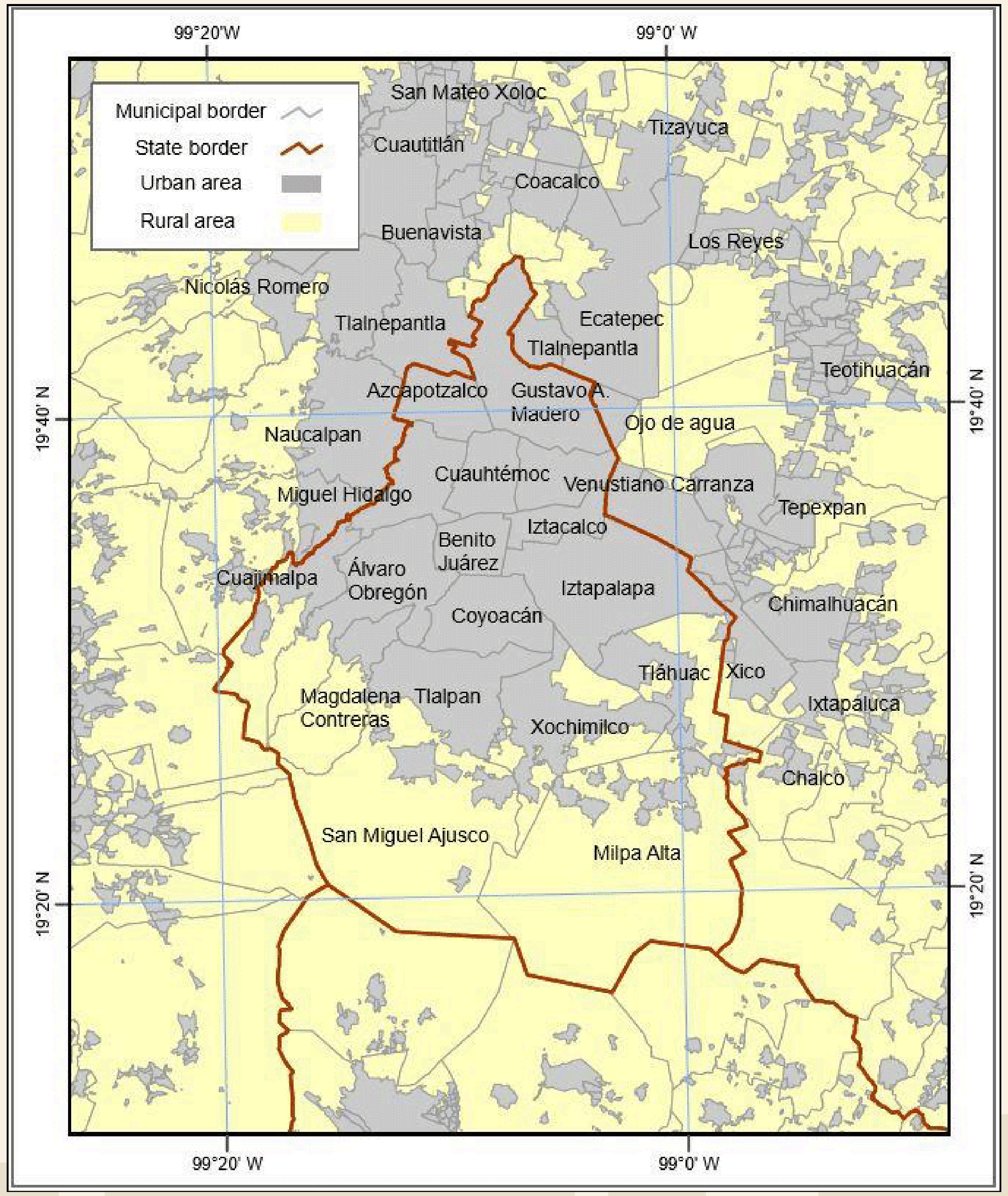

Fig. 1. Mexico City & Metropolitan Area of the Valley of Mexico (based on data from INEGI 2016). (Wieschalla M. N., 2016)

READING THE MAPS: H VS H′ AND THE RIDGE PATTERN

1. H (geopotential height): height of the 500 hPa surface.
2. H′ (anomaly): deviation from the daily climatology (positive = "thicker/warmer," negative = "thinner/cooler").
3. Ridge (dorsal): warm "bubble" aloft → subsidence, stability, weak ventilation.
4. Vectors: 500 hPa wind toward direction (small magnitudes on peak days).
5. Key takeaway: Positive H′ + weak flow = subsidence cap → pollution accumulates.

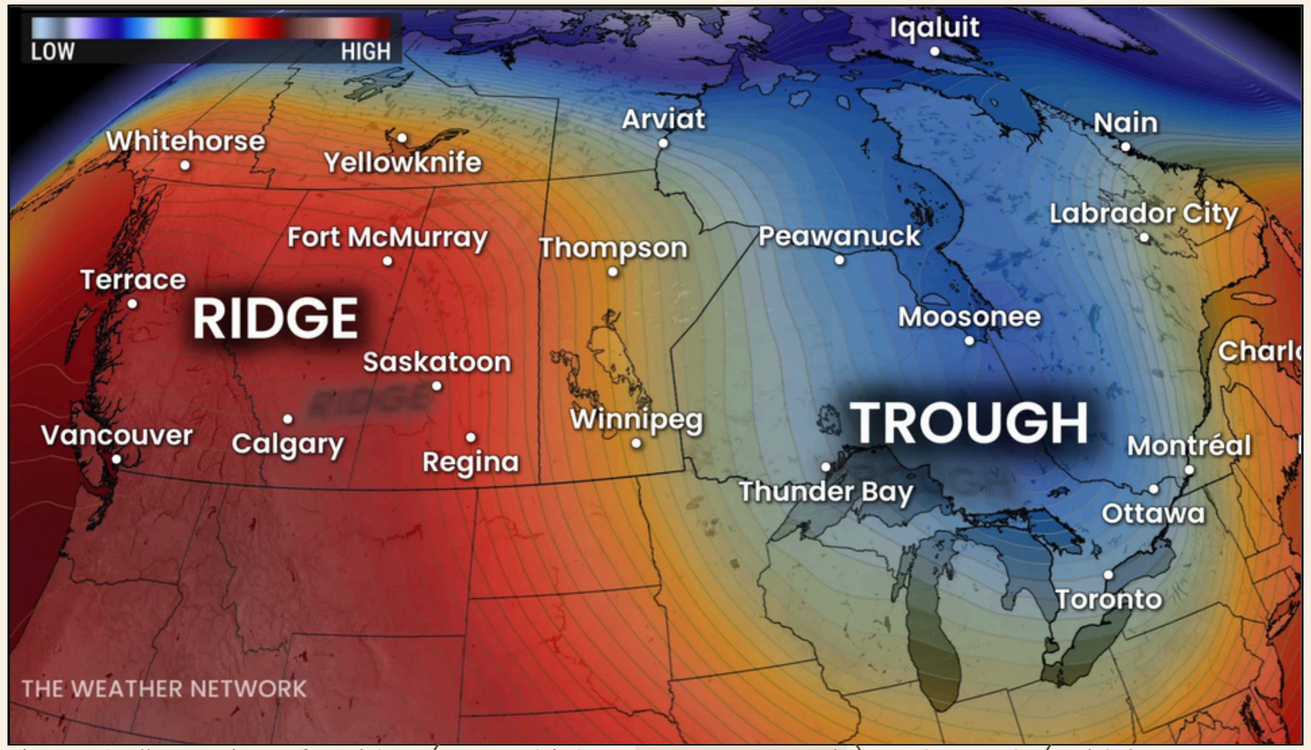

Figure 2. Illustration of a ridge (warm, high-pressure anomaly) and trough (cold, low-pressure anomaly). Ridges bring subsidence and stability; troughs favor lift and ventilation. (The Weather Network, 2025).

500 hPa
PATTERNS

MULTIPANEL OVERVIEW

All pollutants show the same ridge–stagnation fingerprint

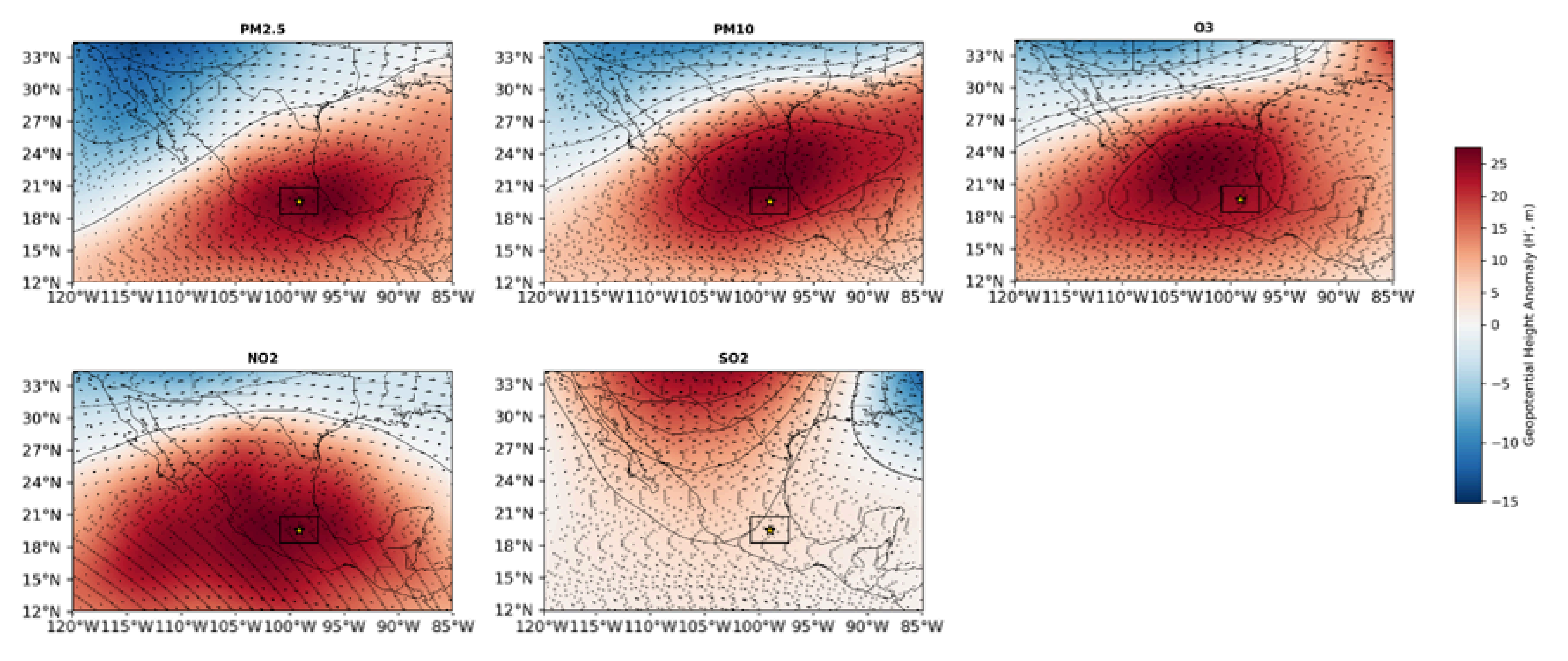

Fig. 3. Multipanel (500 hPa): Composites for $PM_{2.5}$, PM_{10}, O_3, NO_2, SO_2 showing a coherent ridge–stagnation pattern (H′>0) and weak ENE flow during high-pollution days. Stippling marks p<0.05. (Created by the author).

- Positive H′ over central Mexico across PM2.5, PM10, O3, NO2, SO2.
- Very weak 500 hPa flow (≈0.7–1.7 m s^{-1}) oriented toward ENE.
- Stippling marks areas where H′ anomalies are significant ($p < 0.05$).

The synoptic control is coherent across pollutants.

PM2.5 COMPOSITE

PM2.5 - 500 hPa H′ composite

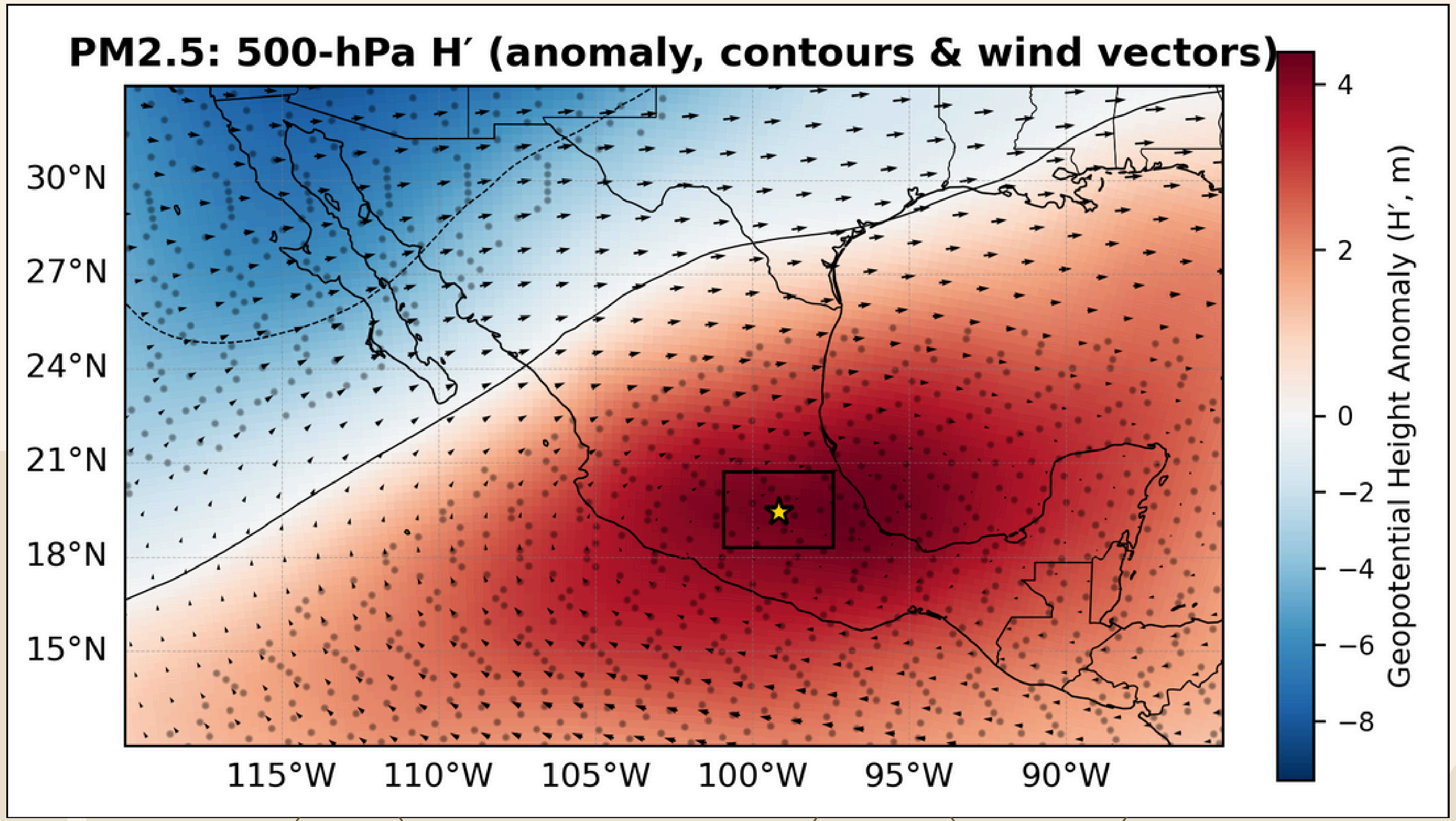

Fig. 4. PM2.5 composite (500 hPa): Geopotential-height anomaly H′ (shading, m), H′ contours (solid = positive; dashed = negative), and 500-hPa wind vectors. An anomalous ridge (H′>0) and weak ENE flow indicate subsidence and stagnation conducive to fine-particle build-up in the MCMA. (Created by the author).

- Broad positive H′ centered near MCMA.
- Weak ENE flow (~1.7 m/s) above the basin → low ventilation.
- Consistent with fine-particle build-up under warm, clear, subsident conditions.

Subtropical ridge days strongly favor PM2.5 accumulation.

PM10 COMPOSITE

PM10 - strongest ridge signal

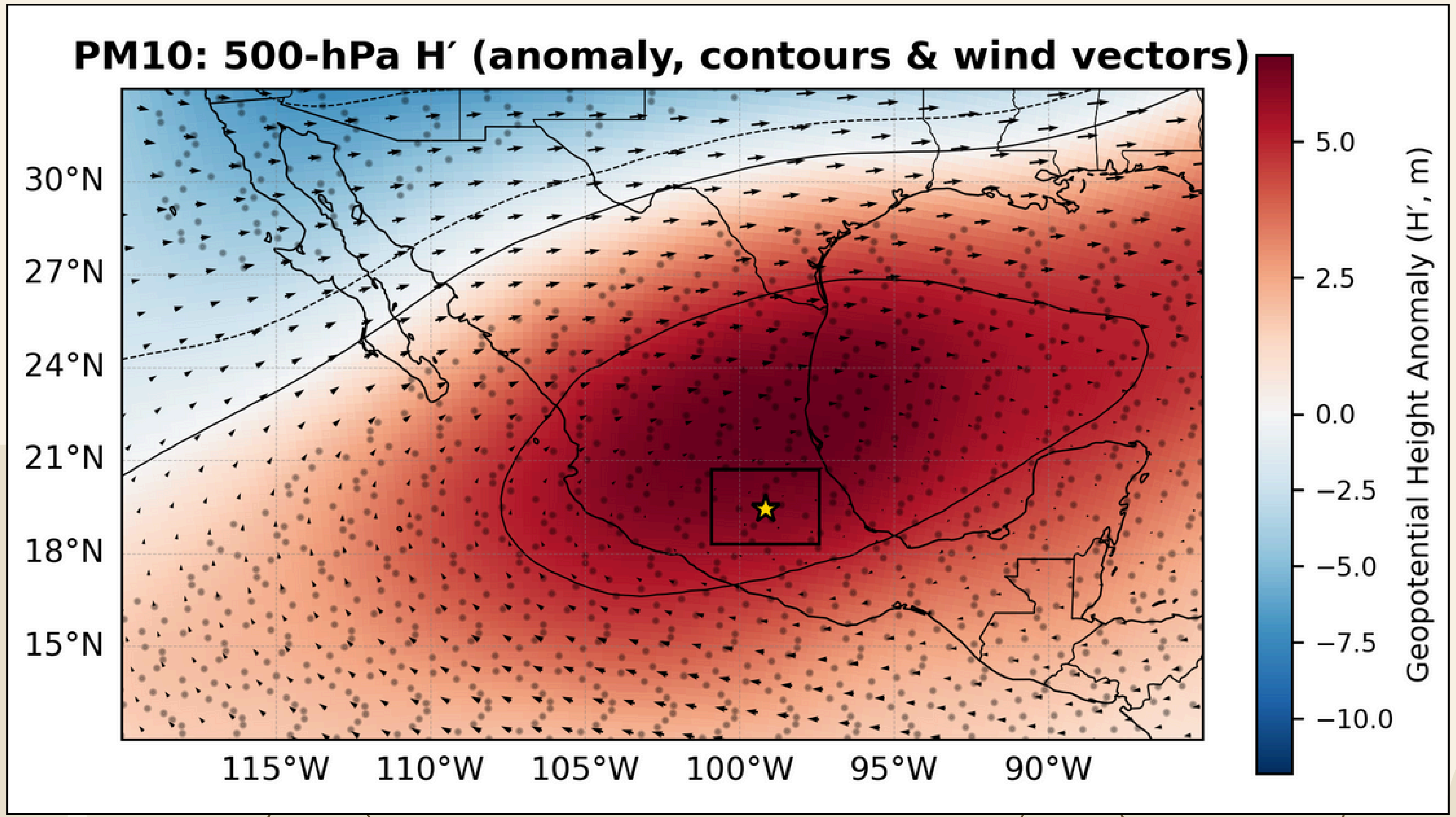

Fig. 5. PM10 composite (500 hPa): Same as Fig. 4. Strongest ridge signal across pollutants (H′≈+6 m) with very weak NE/ENE flow, a classic dry-stagnation pattern favoring coarse particles. (Created by the author).

- Largest H′ (~+6 m) among pollutants; closed positive contours over MCMA.
- Very weak NE→ENE flow (~0.9 m/s).
- Dry, stagnant episodes favor coarse particles and re-suspension.

PM10 peak days are textbook "blocking + stagnation."

O3 COMPOSITE

O3 - compact ridge and minimal winds

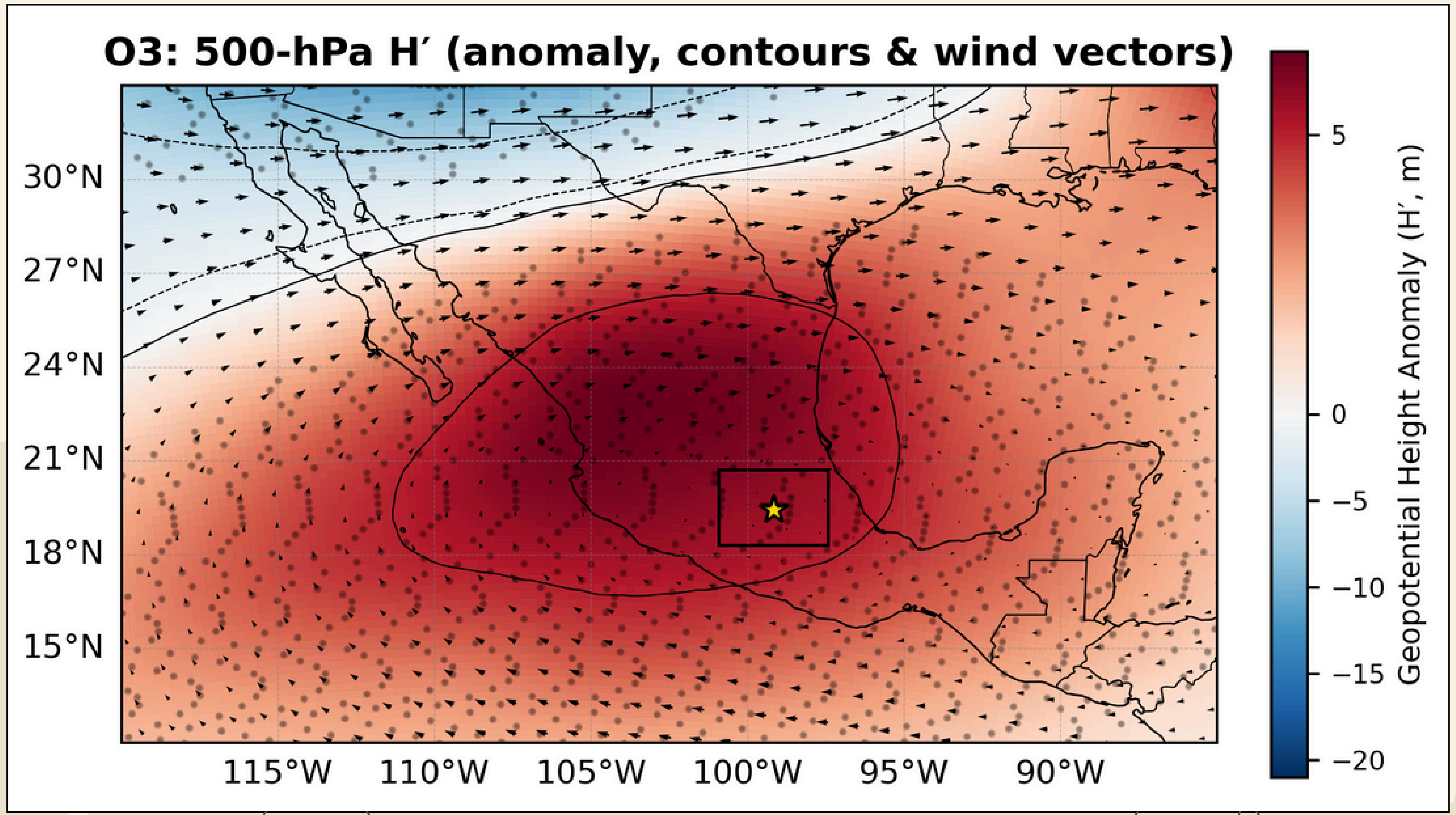

Fig. 6. O3 composite (500 hPa): Same as Fig. 4. Compact positive H′ core and minimal winds (≈0.75 m/s) consistent with subsident, clear-sky conditions that enhance ozone formation and limit dispersion. (Created by the author).

- Compact positive H′ core (~+5.6 m).
- Weakest winds across species (~0.75 m/s).
- Clear skies + stability → enhanced photochemistry, limited dispersion.

Ozone peaks align with warm, subsident, low-wind days.

NO2 COMPOSITE

NO2 - primary-pollutant stagnation

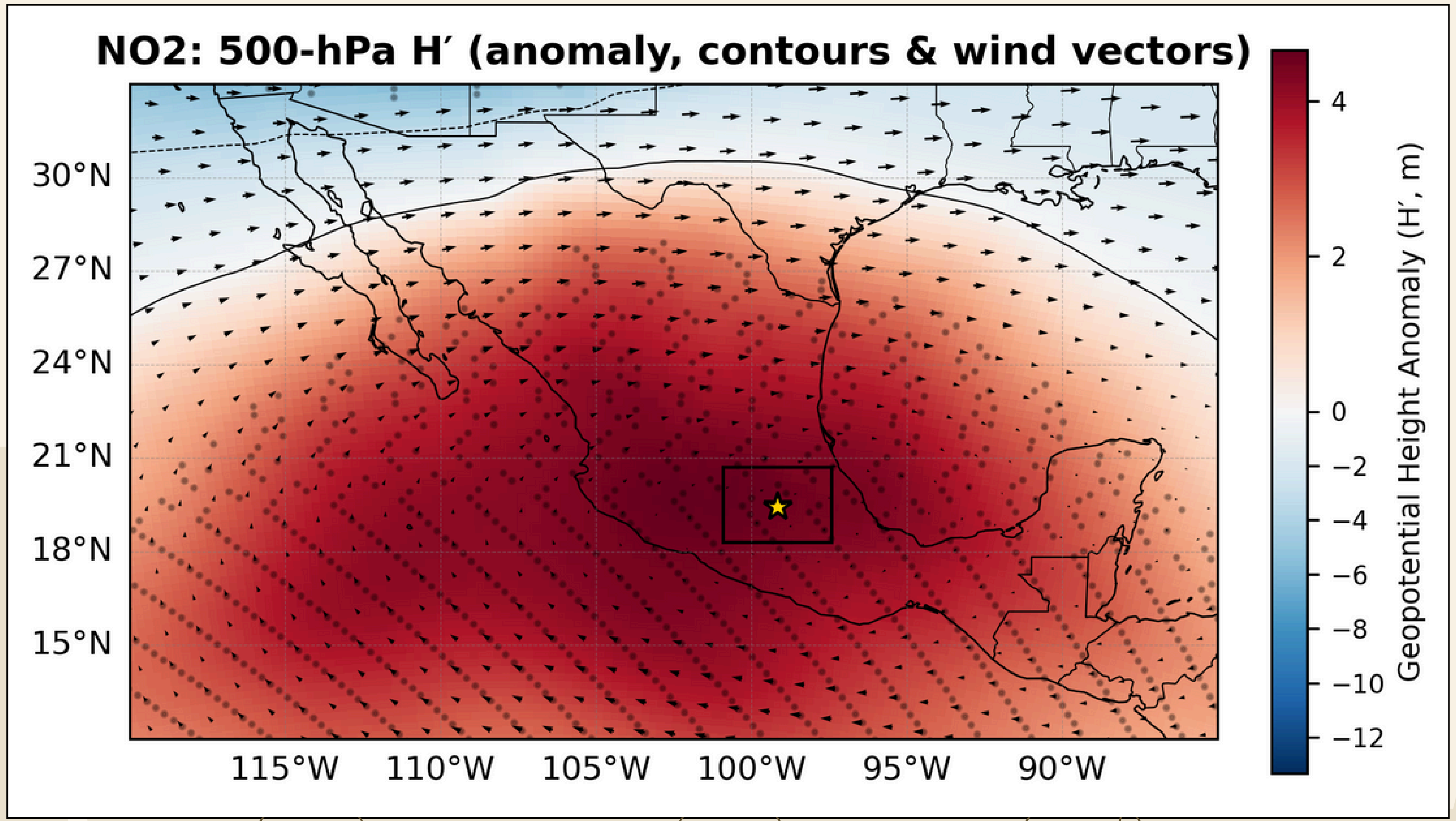

Fig. 7. NO2 composite (500 hPa): Same as Fig. 4. Positive H′ (~+4.5 m) and weak E-NE flow (~1.35 m/s) imply reduced mixing and advection, favoring primary-pollutant accumulation. (Created by the author).

- Basin-scale ridge (H′ ~ +4.5 m).
- Light E–NE flow (~1.35 m/s).
- Weak vertical mixing → near-surface accumulation inside the valley.

Anticyclonic stability suppresses NO2 ventilation.

SO2 COMPOSITE

SO2 - broader ridge footprint

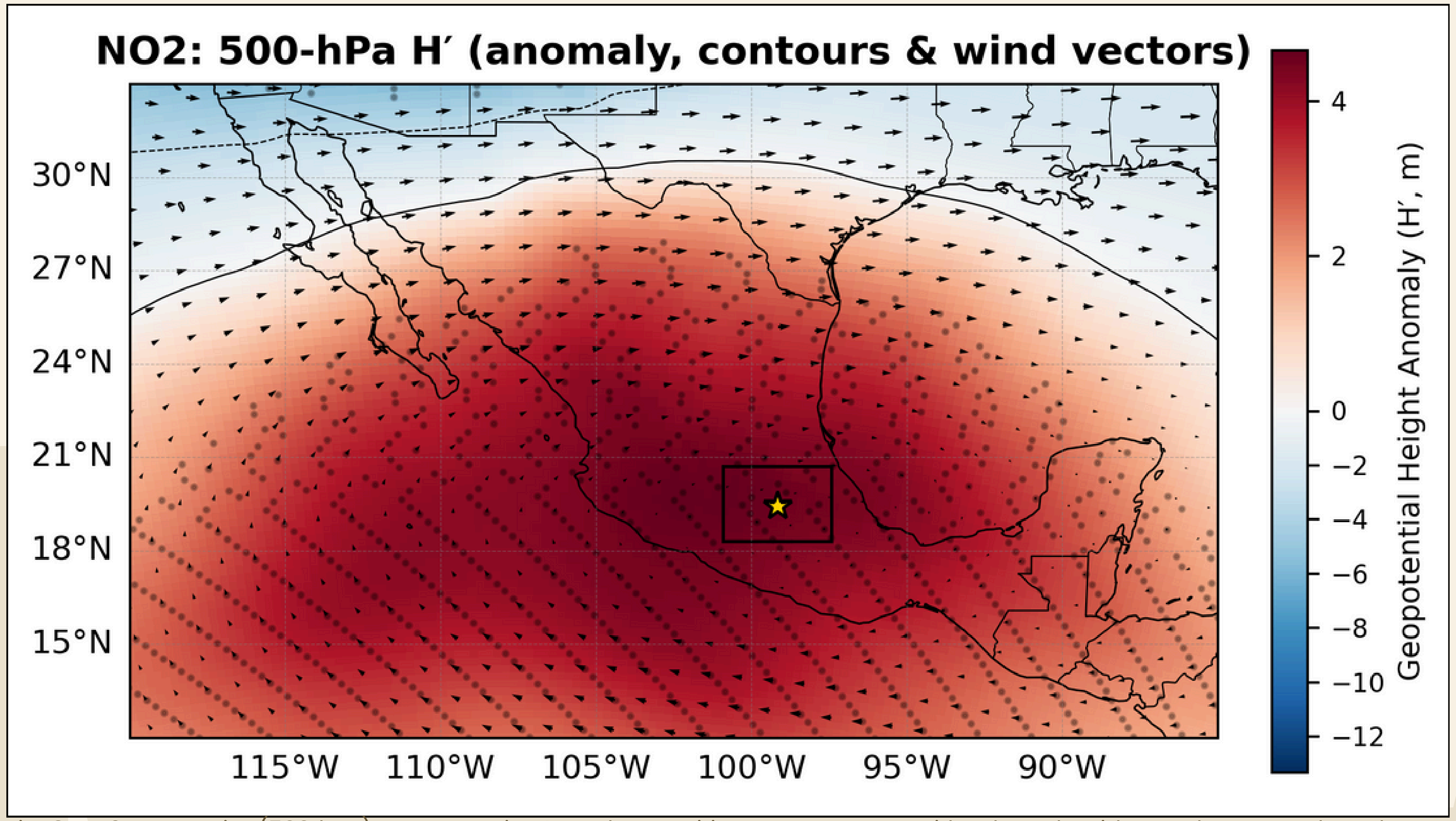

Fig. 8. SO2 composite (500 hPa): Same as Fig. 4. Modest positive H′ over MCMA with a broader ridge to the NW and weak E–NE winds (~0.8 m/s), indicating regional-scale subsidence and limited ventilation. (Created by the author).

- Positive H′ over MCMA with ridge limb extending NW.
- Very weak E–NE flow (~0.8 m s^{-1}).
- Suggests regional-scale subsidence › tight local core.

Same mechanism, broader spatial control.

QUANTITATIVE SUMMARY

Mean anomaly and flow over MCMA on peak days

Table 1. Mean mid-tropospheric anomaly and flow during high-pollution days in Mexico City (2012–2024). H′ is the 500-hPa geopotential-height anomaly averaged over the MCMA box; winds are 500-hPa mean composites over the same box. Azimuth is the direction toward which the composite vector points (clockwise from north), indicating weak ENE flow (<2 m/s) across pollutants. (Created by the author).

Pollutant	Mean H′ (m)	Mean wind (m/s)	Azimuth (°)	Direction (toward)
PM2.5	4.17	1.7	61	ENE
PM10	6.1	0.91	43	NE
O3	5.58	0.75	56	E-NE
NO2	4.51	1.35	72	ENE
SO2	4.44	0.78	71	ENE

A +4–6 m H′ ridge and calm ENE flow characterize all peak days.

MONTHLY DISTRIBUTION

When do peak days happen? (p90 per month, 2012–2024)

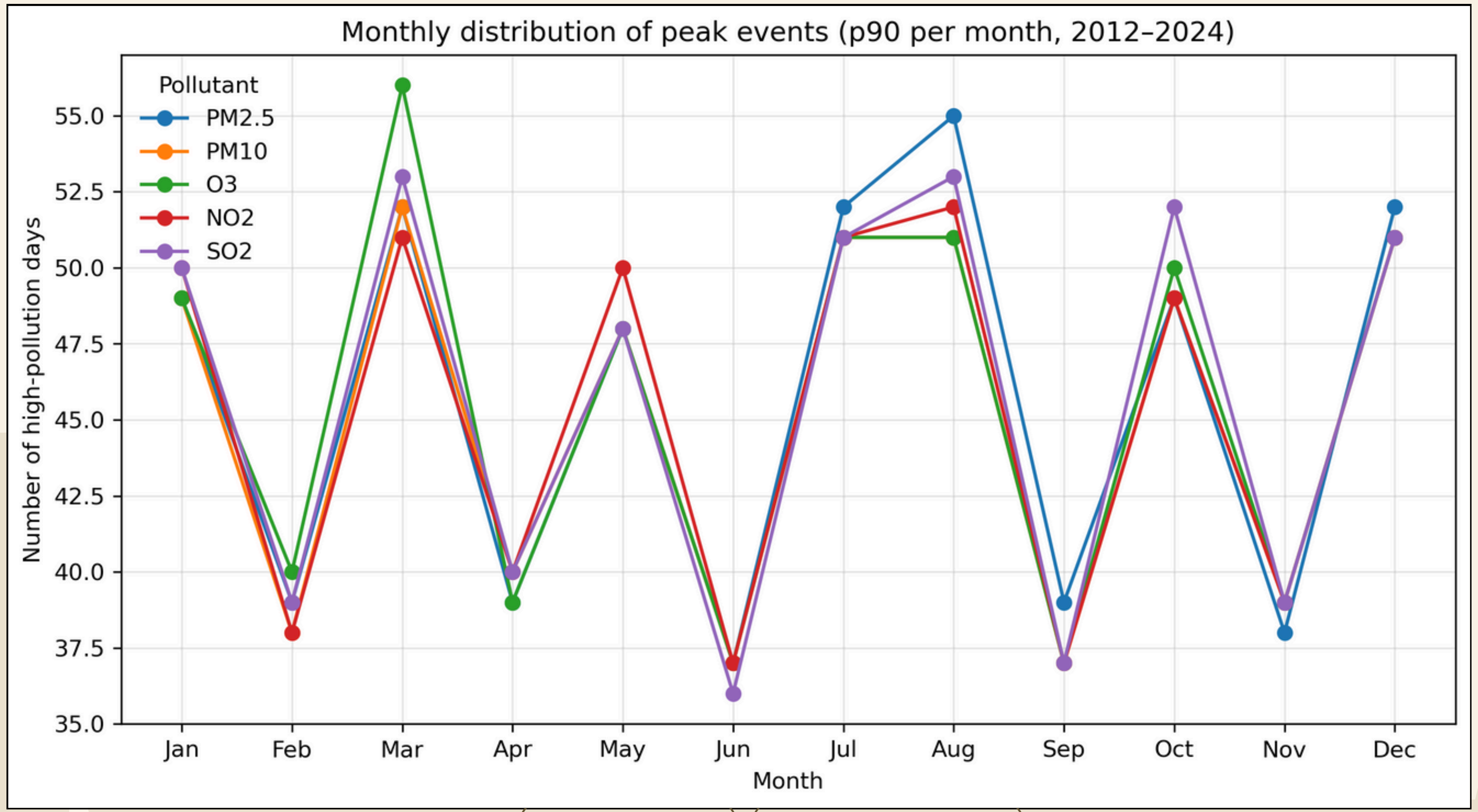

Fig. 9. Monthly counts of high-pollution days (p90 within month). (Created by the author).

- Bimodal seasonality: March and Jul–Aug maxima.
- Minima in Jun/Sep (rainy-season convection and ventilation).
- Autumn rebound (Oct–Dec) consistent with stable post-rainy conditions.

Seasonality tracks the ridge–stability cycle.

SIDE-BY-SIDE

RESULTS VS. MX LITERATURE

OWN RESULTS VS. MEXICO LITERATURE (1/2)

Alignment with Mexican studies

- *Díaz-Esteban et al., 2022*:
  - Highest pollution under mid-tropospheric ridge over central MX (subsidence, warm/dry, weak winds) → matches ridge/stability pattern.

- *Silva-Quiroz et al., 2019*:
  - Blocking highs (>500-hPa positive anomalies) on ozone contingency days → same anticyclonic anomaly and stagnation.

The composites replicate the classic ridge/blocking regimes.

OWN RESULTS VS. MEXICO LITERATURE (2/2)

Surface-level consistency

- *Salcido et al., 2019*:
  - Weak/recirculating surface/local winds and basin trapping on polluted days → matches weak upper-level winds and poor ventilation.

- *Jáuregui & Luyando, 1992*:
  - Anticyclones favor nocturnal drainage + daytime up-valley recirculation → stagnation → consistent with weak 500 hPa ridge control.

Upper-level ridge pairs with local basin recirculation at the surface.

FACE-TO-FACE

HK LITERATURE VS. RESULTS/MX LITERATURE

HONG KONG ATMOSPHERIC REGIMES

Synoptic and local drivers of pollution in Hong Kong

HK's pollution episodes arise from a mix of regional inflow (monsoon/TC) and local sea–land-breeze trapping under subsidence.

- Winter monsoon & trans-boundary transport → Mainland inflow under N/NE winds (Luo 2018).
- Tropical-cyclone modulation:
  - Eastern TCs: raise $PM_{2.5}$ via subsidence + northerlies.
  - Western TCs: often ventilate the region (Zeng 2025).
- Sea–land-breeze recirculation: daytime convergence + nocturnal stability trap pollutants locally (Liu & Chan 2002).
- Multi-pollutant coupling: PM–O_3–NO_2 interactions strongest during stagnant, post-TC subsidence (Lu & Wang 2004).

MEXICO CITY VS. HONG KONG COMPARISON

CDMX vs. HK: Common physics, different forcing

- Shared mechanism: subsidence + weak winds → stagnation and pollutant build-up.
- Differences:
  - *Forcing*: CDMX = persistent mid-tropospheric ridge; HK = monsoon & tropical-cyclone modulation.
  - *Source*: CDMX = local accumulation; HK = strong trans-boundary import.
  - *Boundary layer*: CDMX = mountain-basin inversions; HK = sea-breeze recirculation.
- Visually: ridge (500 hPa) vs cyclone (N-E flow).

Table 2. Summary of main synoptic differences between CDMX and HK. (Created by the author).

City	Main synoptic driver	Typical flow	Key pollution process
CDMX	500 hPa ridge / subsidence	ENE < 2 m s^{-1}	Stability + basin trapping
HK	Monsoon / TC subsidence	N–NE > 4 m s^{-1}	Transport + recirculation

CDMX peaks are home-grown stagnation under a ridge, while HK peaks are imported and cyclone-modulated.

SUMMING UP

CONCLUSIONS + NEXT STEPS

SYNTHESIS & "THE PATTERN"

What this means for Mexico City

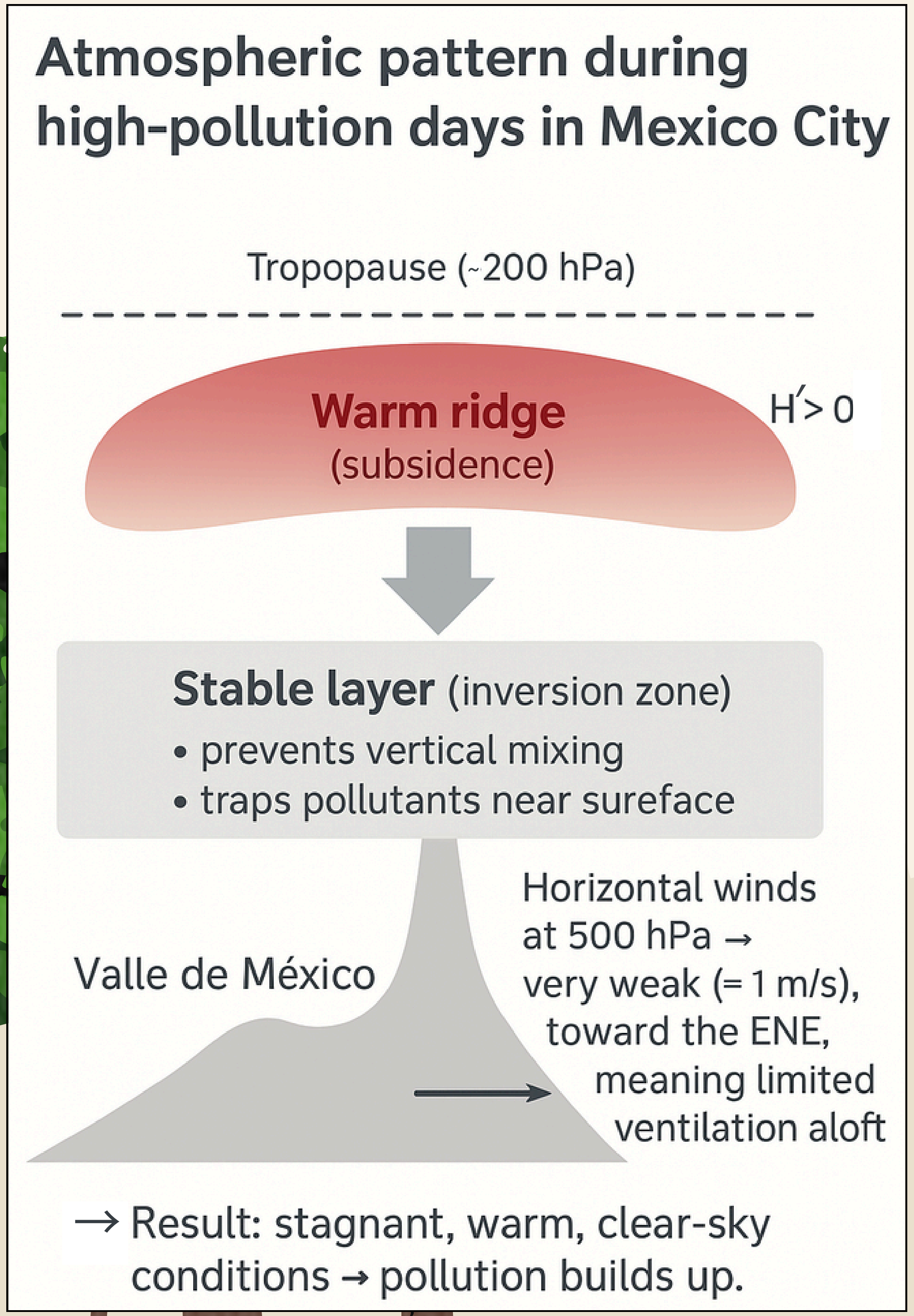

Fig. 10. Simplified schematic of the ridge-stagnation mechanism over the Mexico City basin: warm subsiding air aloft caps the cooler polluted layer below, limiting vertical mixing. (Created by the author using Adobe Firefly).

- Diagnostic fingerprint: H′>0 ridge over central MX + weak ENE flow at 500 hPa.
- Physical chain: ridge → subsidence & stability → shallow mixing layer → low ventilation → pollutant build-up.
- Seasonality: Spring & late-summer peaks; rainy-season minima.

The ridge–stagnation regime is the primary synoptic driver of CDMX high-pollution days.

NEXT STEPS

- Write the scientific draft report (Abstract → Introduction → Methodology → Results & Discussion → Conclusions & Recommendations; + References/Appendices). Focus on Week-6 insights; summarize Weeks 1–5 in Methodology.
- Prepare the final draft presentation slides focusing on the whole "journey" (weeks 1-6).

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THANK YOU!

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